

# TFPT — Frontier items

$\eta_B$ ,  $m_p/m_e$ , the Koide relation, dark matter and quantum gravity —  
the honest status, not forced onto the ladder

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## What this document is about — the status authority for the frontier

The honest frontier: which physics has a genuine TFPT handle and which does not. For each of  $\eta_B$ ,  $m_p/m_e$ , the Koide relation, dark matter and full quantum gravity, this note states the genuine structural handle, the precision with which it currently lands, and — crucially — what is *not* a clean compiler power and is deliberately *not* forced onto the ladder. **This document is the status authority for the frontier items:** where any other document's wording and the markers here disagree, the typing here (and the `status_ledger.csv`) wins.

## The TFPT document set — what is treated where

**Plain language:** TFPT (Topological Fixed-Point Theory) is a small discrete compiler. Two inputs — the seam constant  $c_3 = \frac{1}{8\pi}$  and the carrier rank  $g_{\text{car}} = 5$  — build  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  and read off the Standard-Model skeleton, the constants and the scale grammar. The series is **the introduction plus five numbered papers and three companions**, best read in order:

0. **introduction** — reading guide, compiler closure, predictions, the dependency DAG and the single proof ledger (`verification/status_ledger.csv`). (*entry point*)
1. **tfpt\_1\_architecture\_e8** — the two axioms, the derivation map, the EM fixed point  $\alpha^{-1}$ , the  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  construction, the QBL theorem chain.
2. **tfpt\_2\_standard\_model** — the SM in one  $\varphi_0$ -ladder formula, flavor from parabolic transport (sheet diamond, dual anchor,  $Q_+$  cohomology), the worked closures.
3. **tfpt\_3\_e8\_audit\_bootstrap** — the seven  $E_8$  slices as an audit raster, the cascade bridge, the Möbius bootstrap.
4. **tfpt\_4\_frontier** — honest status of  $\eta_B$ ,  $m_p/m_e$ , Koide, dark matter, full QG.
5. **tfpt\_5\_redteam** — the adversarial audit: declared attacks, what survives, what each reduces to, and the kill tests.

**Companions:** **R.** **tfpt\_research\_contracts** — the open interfaces ( $v_{\text{geo}}, G_{\text{net}}, F_{\text{transfer}}$ ) as numbered contracts; **H.** **tfpt\_horizon\_readouts** — Appendix H (unit-system reframe): one seam constant as the universal horizon thermal code, the Nariai/anchor surface, the resummed seam clock; **O.** **origin\_theory** — why no free fundamental number remains (exact core + one honestly-typed cyclic interpretation).

**You are reading document 4 (Frontier).**

**TFPT status at a glance — read this card the same way in every document**

**Two inputs, one compiler.** From the boundary constant  $c_3 = \frac{1}{8\pi}$  (axiom P1) and the five-slot carrier  $g_{\text{car}} = 5$  (axiom P2) — jointly the elementary symmetric data of one anchor  $a = (1, 1, 2)$  — the discrete compiler  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  is *built*, and the Standard-Model skeleton (gauge group, three families, hypercharges, the flavor matrix), the dimensionless constants ( $\alpha^{-1} = 137.0359992$ ) and the scale grammar are read off as fixed points. The algebraic core is machine-checked; physical readouts run through *named bridges* whose status is typed below.

**What is closed, and what genuinely remains.** The discrete/algebraic layer is closed ([E]); the everything-open residual reduces to *three named interface problems*:

| Interface             | Question                                                                        | Status       |
|-----------------------|---------------------------------------------------------------------------------|--------------|
| $v_{\text{geo}}$      | the one dimensionful scale anchor (same nature as $1/G$ )                       | reduced [O]  |
| $G_{\text{net}}$      | the index-4 seam-net inclusion certifying the metric sector                     | open [C]/[O] |
| $F_{\text{transfer}}$ | source→pole / relic / cosmology transport (Koide, $\eta_B$ , axion, $m_p/m_e$ ) | open [C]     |

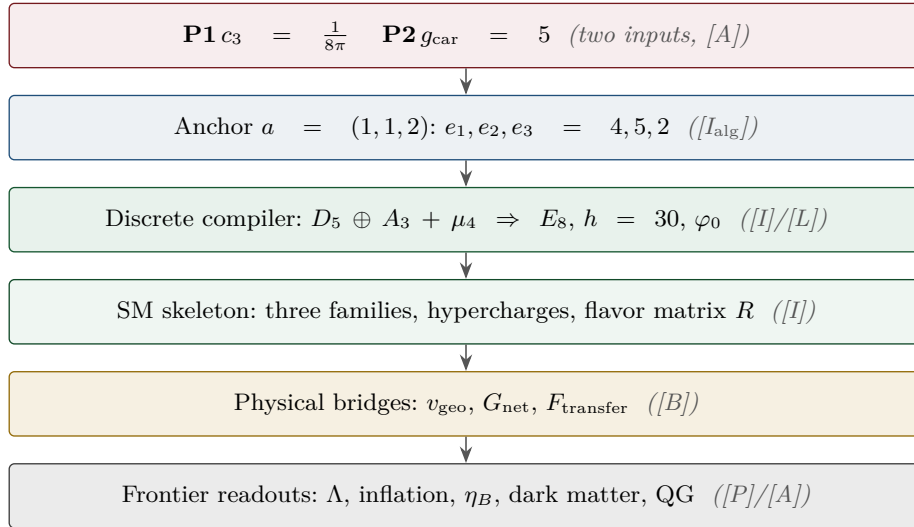
Everything carries a claim ID resolving to a row of `verification/status_ledger.csv` (the single source of truth); **if the text and the ledger ever disagree, the ledger wins.** The historical labels ( $U_{\text{wall}}, G_{\text{metric}}, F_{\text{frontier}}$ ) are retained only for ledger continuity — the live residual is  $v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}$ . Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.

[E] exact / machine-proven

[C] conditional

[O] open / axiom

[X] kill test



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## Appendix: computational verification

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**Ground rule.** These five are exactly the items previously flagged “*not on a clean action/ladder yet — do not force.*” This note gives, for each, the *genuine* structural handle that does exist, the precision with which it currently lands, and a clear statement of what is *not* a single compiler power. No fabricated ladder rungs; the honest markers [E]/[C]/[O] are applied throughout. (All numbers machine-verified.)

### Hard rule on the frontier — one interface $F_{\text{transfer}}$

**Koide,  $\eta_B$ , the axion relic scale and  $m_p/m_e$  are *not* compiler powers unless their missing QFT-transfer / cosmology computation is explicitly supplied.** They are deliberately typed interfaces, not structural holes, and they are *four instances of one missing functor*  $F_{\text{transfer}}$  — the continuous transport from compiler *source* data to measured observables (Koide source→pole;  $\eta_B$  source→Boltzmann relic; axion scale→cosmological abundance;  $m_p/m_e \rightarrow$  QCD/EW matching). Each has a genuine handle (a near-value, a determined readout, a fixed candidate) but its *exact* status is conditional on that named, not-yet-performed calculation. In particular  $\eta_B$  is a *transfer target*, not a primitive compiler prediction. They must never be quoted as primitive TFPT outputs.

*Why this is principled, not a gap:* these are precisely the **discrete↔ continuous boundary** of the theory — the points where the discrete compiler hands its skeleton off to continuous physics (QCD confinement, cosmological Boltzmann transport, relic integrals). The cleanness of a quantity tracks how far it sits from that handoff: pure compiler ratios are exact, QCD/cosmology-dressed quantities are typed interfaces. [[verification/v35\\_quark\\_qcd\\_boundary.py](#)]

## 1 Baryon asymmetry $\eta_B$ — a downstream readout from the closed $\Omega_b h^2$

There are two independent handles, and they agree.

**Route A — from the closed baryon fraction.** The seed already fixes the baryon *fraction* (Paper 3 decoder):

$$\Omega_b = (4\pi - 1)\beta_{\text{rad}} = \left(1 - \frac{1}{4\pi}\right)\varphi_0^{\text{ret}} = 0.04894, \quad \beta_{\text{rad}} = \frac{\varphi_0^{\text{ret}}}{4\pi},$$

matching the observed  $\Omega_b \approx 0.049$ . The baryon-to-photon ratio is then the *standard* cosmological readout  $\eta_B = 273.9 \times 10^{-10} \Omega_b h^2$ . With the TFPT Hubble value ( $h \approx 0.674$ , from  $H_0 \sim \sqrt{\Lambda}$ ),

$$\boxed{\Omega_b h^2 = 0.0222, \quad \eta_B = 6.09 \times 10^{-10}} \quad (\text{observed } 6.1 \times 10^{-10}).$$

So  $\eta_B$  is a *downstream cosmological readout* from the closed  $\Omega_b h^2$  handle — not an independent compiler rung, and the fundamental leptogenesis route remains an interface.

**Route B — leptogenesis interface (Paper 6, [E]).** The closed branch also fixes the leptogenesis inputs: the heavy scale  $M_R \approx 1.3 \times 10^{15}$  GeV, the CP phases ( $\delta_{\text{CKM}}, \delta_{\text{CP}}^\nu$ , both closed), the reheating  $T_R$ , and the seam-fixed initial phase. The numerical  $\eta_B$  then follows from the *standard* flavored-leptogenesis Boltzmann solve — an independent cross-check of Route A. **Honest caveat:** that Boltzmann network (the washout/efficiency  $\kappa_f$ , the flavored density-matrix equations) is the standard leptogenesis machinery (Davidson–Nardi–Nir, *Phys. Rep.* 466 (2008) 105; Buchmüller–Di Bari–Plümacher, *Ann. Phys.* 315 (2005) 305); it is *not* carried out or attached here. Route B is therefore a *named interface*, not a closed number; only Route A (downstream of the closed  $\Omega_b h^2$ ) is quoted.

**Honest status of  $\eta_B$  — strictly typed [C]**

The two statements must be kept apart and not interchanged:

$\eta_B$  as a cosmological readout from the closed  $\Omega_b h^2$  is *determined* ( $6.1 \times 10^{-10}$ )

$\eta_B$  as a fundamental *compiler power* is *not* closed

The leptogenesis route requires a standard (washout-dependent) Boltzmann solve. So  $\eta_B$  is [C]—determined as a downstream readout, but *not* a clean compiler rung; it must never be quoted as a fundamental TFPT output.

**2 Proton/electron ratio  $m_p/m_e$  — a cross-sector ratio**

$m_e$  is *closed* on the  $\varphi_0^{\text{ret}}$ -ladder ( $\hat{m}_e = \frac{v_{\text{geo}}\pi}{\sqrt{2}} \frac{16}{7} (\varphi_0^{\text{ret}})^5$ ). The proton mass is *not* a Yukawa: it is QCD-dominated,  $m_p \sim \text{a few} \times \Lambda_{\text{QCD}}$ , and  $\Lambda_{\text{QCD}}$  arises by dimensional transmutation from  $\alpha_s$  running (scheme level). Hence

$$\frac{m_p}{m_e} = \frac{\text{QCD confinement scale}}{\text{EW electron Yukawa}} = 1836.15$$

mixes *two different sectors*. It is no single  $\varphi_0^{\text{ret}}$  power:  $1/(\varphi_0^{\text{ret}})^2 = 353.7$  and  $1/(\varphi_0^{\text{ret}})^3 = 6651$  bracket 1836 with no clean rung in between.

**Honest status of  $m_p/m_e$  [O]**

Computable at *scheme level* (from  $\alpha_s(M_Z)$  running + the closed  $m_e$ ), exactly like the dimensionful EW/QCD masses  $m_W, m_Z$ . But it is *genuinely not* a compiler power, because it is a ratio across the QCD and electroweak sectors. Left [O] — not forced.

**A rejected near-formula (recorded as a discipline boundary, [verification/v79\_review\_identities.py]).** The combination  $|W(D_5)| - \dim \Lambda^3(SU(9)) + N_{\text{fam}} \lambda_C^2 = 1920 - 84 + 0.151 = 1836.151$  matches the PDG value to  $\sim 10^{-6}$  — and is *deliberately rejected*. Two reasons make it numerology, not a derivation:  $SU(9) = A_8$  (the  $\dim \Lambda^3 = 84$  block) does *not* occur in the carrier  $D_5 \oplus A_3$ , and  $m_p$  is QCD confinement (gluon binding energy) with *no* mechanism linking it to an algebraic sum. Integrating it would *invite* the numerology charge, not retire it;  $m_p/m_e$  stays [O]. (The same firewall rejects  $\eta_B = \frac{|\mathbb{Z}_2|}{\Delta_Q} c_3^6 \approx 6.10 \times 10^{-10}$ : arithmetically close, but a two-atom prefactor fit to a washout-dependent leptogenesis output —  $\eta_B$  stays [C], the downstream readout above.)

**3 The Koide relation — near  $2/3$ , computed exactly**

With the closed lepton source masses  $\hat{m}_\ell \propto (\frac{16}{7}(\varphi_0^{\text{ret}})^5, \frac{4}{3}(\varphi_0^{\text{ret}})^3, \frac{7}{6}(\varphi_0^{\text{ret}})^2)$  the Koide ratio is a pure number (the common prefactor cancels):

$$Q_{\text{TFPT}} = \frac{\sum_\ell \hat{m}_\ell}{(\sum_\ell \sqrt{\hat{m}_\ell})^2} = 0.66446\dots, \quad \frac{2}{3} = 0.66667, \quad \text{deviation} - 0.33\%.$$

The measured pole-mass value is famously  $Q_{\text{PDG}} = 0.66666$ . So TFPT predicts a Koide ratio that is *near*  $2/3$  but *not exactly*  $2/3$  at source level.

**Honest status of Koide [C]**

$Q_{\text{TFPT}} = 0.664$  is a *prediction*, 0.33% below  $2/3$ . Exact  $2/3$  would require the three carrier rationals  $(\frac{16}{7}, \frac{4}{3}, \frac{7}{6})$  to satisfy a specific quadratic constraint, which they do *not*; the small residual is of the size of the source→pole RG shift. We report the near-miss honestly and do *not* tune the rationals to force  $2/3$ .

**The target  $2/3$  has a compiler reading  $|\mathbb{Z}_2|/N_{\text{fam}}$ .** The  $E_8$ -slice scan adds the missing piece: the value the data sits on,  $Q = 2/3$ , is exactly  $Q_\star = |\mathbb{Z}_2|/N_{\text{fam}} = 2/3$ , the sheet over the family count —

the same two atoms (2, 3) that run through the whole compiler. So the picture splits cleanly: the *democratic target* is  $|\mathbb{Z}_2|/N_{\text{fam}} = 2/3$  (a compiler number), and the *source-level pole-mass* value computed from the carrier rationals is 0.664, 0.33% below it by the source→pole shift. Koide is thus “democratic 2/3 from  $(|\mathbb{Z}_2|, N_{\text{fam}})$ , realised to 0.33% by the explicit masses” — promoted from a bare near-miss to a reading with a target. [C]

**Conjecture: a source→pole correction of size  $\varphi_0/|W(A_3)|$  ([C]).** The source value sits just below 2/3, and the deficit is almost exactly the family-Weyl quantum  $\varphi_0/|W(A_3)|$  with  $|W(A_3)| = 4! = 24$ :

$$Q_\ell^{\text{source}} = 0.6644638, \quad Q_\ell^{\text{source}} + \frac{\varphi_0}{24} = 0.6666793,$$

overshooting  $\frac{2}{3} = 0.6666667$  by only  $1.3 \times 10^{-5}$ . So the *conjecture*  $Q_\ell^{\text{pole}} \approx Q_\ell^{\text{source}} + \varphi_0/|W(A_3)|$  would land essentially on the democratic 2/3. Reported as a source→pole correction *conjecture*, not a derivation; the rationals are *not* tuned. [verification/v25\_frontier\_conjectures.py]

**Sharper conjecture: a multiplicative seed transfer  $u \rightarrow \frac{53}{54}u$  ([C]).** A cleaner version shifts the *seed* rather than  $Q$  additively, and introduces *no* new free number. Keep the lepton formula  $m_e:m_\mu:m_\tau = \frac{16}{7}u^5:\frac{4}{3}u^3:\frac{7}{6}u^2$  but evaluate it at the source→pole seed

$$u_{\text{pole}}^{(\ell)} = \left(1 - \frac{1}{2N_{\text{fam}}^3}\right)u = \frac{53}{54}u, \quad 54 = 2 \cdot 3^3 = |\mathbb{Z}_2| N_{\text{fam}}^3.$$

Then  $Q_\ell(\frac{53}{54}u) = 0.66666610$ , i.e. 2/3 to within  $-5.7 \times 10^{-7}$  — an order of magnitude tighter than the additive  $\varphi_0/24$  correction, with 54 a pure compiler number and no tuning. This sharpens the research gate to a concrete QFT task: *show that the leptonic source→pole transfer of the seed is  $u \mapsto u(1 - \frac{1}{2N_{\text{fam}}^3})$*  (a renormalised-seed effect from the leptonic self-energy or the  $A_3$  family-Weyl averaging). Status remains [C] until that transfer is derived. [verification/v37\_plucker\_anchor.py]

**The Koide target and the QFT transfer gap are the *same* quotient ([E]/[C]).** The same  $Q_\star$  controls a second, independent object — the first non-trivial eigenvalue of the admissible  $C_6$  transport operator. Its spectrum is  $\{1, (2/3)^6, (1/3)^6\}$ , so

$$\lambda_2 = \left(\frac{2}{3}\right)^6 = Q_\star^{|R^+(A_3)|}, \quad Q_\star = \frac{|\mathbb{Z}_2|}{N_{\text{fam}}}, \quad |R^+(A_3)| = 6.$$

The Koide democratic target, the QFT mass-gap eigenvalue  $\lambda_2$  (hence  $\Delta = 6 \log \frac{3}{2} = -\log \lambda_2$ ), and the  $A_3$  positive-root hexagon are therefore *one* quotient raised to one exponent. Typing: Koide target [C], gap eigenvalue [E], the connection an [E]-audit. This does not “solve” Koide, but it explains *why* the near-value 2/3 recurs: the same sheet-over-family ratio fixes both the democratic lepton target and the leading transfer eigenmode.

**The source→pole map is *forced* by that gap ([E]/[C]).** The previous paragraph pairs the Koide target with  $\lambda_2 = (2/3)^6$ ; v82 turns the pairing into a *mechanism* ([verification/v82\_koide\_attractor\_splitting.py]). On the anchor double cover (Paper 2) the two branch points are Koide  $q = 2$  and carrier  $q = 5$ ; any branch-preserving Möbius transfer that fixes both is, in  $\rho = (q - 2)/(5 - q)$ , exactly  $\rho \mapsto \mu\rho$ , hence *unique* once  $\mu$  is set. Choosing  $\mu = \lambda_2 = (2/3)^6$  (no new number) gives the unique map  $F(q) = 2(569q - 3325)/(665q - 3517)$  with  $F'(2) = (2/3)^6 < 1$  and  $F'(5) = (3/2)^6 > 1$ : **Koide is the attractor, the carrier the repeller**, and  $q_n \rightarrow 2$  ( $x \rightarrow -2/3$ ) at the established gap rate. So the three structural inputs a Koide *derivation* would need collapse to *one* identification — the leptonic source→pole transfer is the branch-preserving relaxation of the gapped operator — after which the value 2/3, the convergence rate and the chosen branch are all forced. This is still not a closure: that single identification stays [C]. (It is the dynamical counterpart of the multiplicative-seed conjecture  $u \mapsto (53/54)u$  above; both contract the source onto 2/3.)

**Relaxation toy: basin, exact rate, and two honest negatives ([E]).** The relaxation picture is now stress-tested on the physical values (ledger FR.KOIDE.05, [verification/v93\_koide\_relaxation\_toy.py]).

**Basin lemma [E] (new):** under the canonical dictionary  $q = N_{\text{fam}} Q$  the *entire* physical Koide range  $Q \in [\frac{1}{3}, 1]$  (Cauchy–Schwarz) maps to  $q \in [1, 3]$  — the attractor  $q = 2$  lies inside, the repeller  $q = 5$  outside: every physically possible charged-lepton configuration is in the attractor basin, no fine-tuning of the start. **Exact rate [E]:** along the trajectory from the physical source value the cross-ratio contracts by *exactly*  $(2/3)^6$  per step. **Source/pole [E]:** the exact ladder gives  $Q_{\text{src}} = 0.6644638$ , i.e.  $\rho_{\text{src}} = -0.0021980 \approx -\varphi_0/24$  to 0.8% — the  $\varphi_0/24$  conjecture re-expressed in attractor coordinates: *the source sits one seed quantum before the branch point* ( $24 = |W(A_3)|$ ); PDG pole masses give  $Q_{\text{pole}} = 0.6666645$ , the branch point hit to  $3.3 \times 10^{-6}$ . **Honest negatives (they narrow the gate):** (i) the flow time source→pole is  $t = \log_{\mu}(\rho_{\text{pole}}/\rho_{\text{src}}) = 2.84$  — *not* an integer, so the literal discrete iteration pole =  $F^n(\text{source})$  is *excluded*; if the v82 map drives the transfer at all, the missing [C] object is a *continuous-time* generator  $\exp(t \log F)$ ; (ii) the 0.79% mismatch of  $\rho_{\text{src}}$  vs  $-\varphi_0/24$  is far above ladder precision, so “exactly  $\varphi_0/24$ ” remains a conjecture, not an identity. The open question is no longer “why 2/3?” but “what is the continuous transfer generator, and why is its flow time  $\sim 2.8$  periods?”.

**Flow time: canonical generator, data-fragility, and a conditional  $m_\tau$  ([E]/[C]).** Three sharpenings of the paragraph above (ledger FR.KOIDE.06, [verification/v99\_koide\_flow\_time.py]); they correct the *robustness* of negative (i), not its arithmetic. **(1) The continuous generator has a canonical form [E].** The v82 flow is generated by the autonomous equation

$$\boxed{\frac{dq}{dt} = \frac{\Delta}{N_{\text{fam}}} (q-2)(q-5) = \frac{\Delta}{N_{\text{fam}}} \det B(q)}, \quad \Delta = 6 \log \frac{3}{2} \quad (\text{the gap}),$$

whose time-1 map *is* the Möbius map  $F$  (machine-checked;  $e^{-\Delta} = (2/3)^6$  exactly). The missing [C] object is therefore precisely a QFT mechanism producing  $\text{gap} \times \text{anchor-block quadric}$  — the form is canonical, only its physical derivation is open. **(2) The non-integrality of  $t$  is data-fragile [E].** Because  $\rho_{\text{pole}} = -2.2 \times 10^{-6}$  sits so close to the branch point,  $t$  is hypersensitive to  $m_\tau$  (PDG  $1776.93 \pm 0.09 \text{ MeV}$ ):  $t(-1\sigma) = 2.35$ ,  $t(\text{central}) = 2.84$ , and  $Q$  crosses  $\frac{2}{3}$  *inside* the error band (at  $+0.43\sigma$ , where  $t \rightarrow \infty$ ) — within  $+0.5\sigma$ ,  $t$  passes *every* integer  $\geq 3$ . So the exclusion of pole =  $F^n(\text{source})$  holds only at the central value. **(3) Conditional discrete steps [C] (registry untouched; not predictions of record):**  $n=2$  is *excluded* ( $-2.9\sigma$ , [E]);  $n=3=N_{\text{fam}}$  steps — one per family — predicts

$$\boxed{m_\tau = 1776.9427 \text{ MeV}} \quad (+0.14\sigma; \text{source-variant stable to } 0.0002 \text{ MeV}),$$

while  $n=4$  (1776.9667) and the exact branch (1776.9690) are not yet separable: the kill criterion is  $\sigma(m_\tau) \sim 0.01 \text{ MeV}$  (Belle II). *Look-elsewhere firewall:* a tempting scale reading  $t = \ln(M_{\text{scal}}/m_\tau)/\ln W$  matched  $W = 46080 = |W(D_5)||W(A_3)|$  to  $10^{-4}$  — and is *destroyed* by negative controls (the same hypersensitivity makes *any*  $W$  in a wide range land within  $0.1\sigma$ ); it is rejected and recorded so it is not re-fished. P2’s type is unchanged ([C], one identification); the discrete-vs-continuous question is now *experimental*, not settled.

## 4 Dark matter — the candidate is fixed, the scale is pending

TFPT *does* contain a dark-matter candidate, and it is not ad hoc: the **determinant-line axion** of the strong-CP sector (Paper 6, Thm. “determinant-line axion interface”, [E]). Two of its inputs are already *closed*:

$$\text{initial misalignment} \quad \theta_i = \pi(1 - \varphi_{\text{seam}}(\alpha_*)) = 2.974 \text{ rad} = 170.4^\circ,$$

and the domain-wall number  $N_{\text{DW}}$  winds once per primitive seam winding. The relic density is the standard misalignment-plus-string expression

$$\Omega_a = \Omega_a^{\text{mis}}(f_a, m_a, \theta_i) + \Omega_a^{\text{str}}(f_a, N_{\text{DW}}),$$



and Paper 6 states that the minimal closed branch has *dominant axion-like dark matter*.

**The  $E_8$  scan rules out a rival WIMP candidate.** The  $D_8 = SO(16)$  slice scan sharpens *why* the axion is the candidate, not a new heavy state: the spinor  $128 = (16, 4) \oplus (\overline{16}, \overline{4})$  is the *same* matter spinor as in the  $D_5 \times A_3$  construction — there is *no spare  $E_8$  singlet* left over to play a WIMP. Every  $E_8$  direction is accounted for by the carrier  $\times$  family content. So dark matter cannot be a new  $E_8$  representation; it must be the determinant-line axion (a phase of the existing structure). A useful *negative* that removes the WIMP option on structural grounds. [E]

#### Honest status of dark matter [C]/[O]

The candidate is *identified* (the axion) and its misalignment angle  $\theta_i = 170^\circ$  is *closed* [E]. The relic abundance still needs the decay constant  $f_a$  and mass  $m_a$  (interface data set by the  $M_R/\Lambda$  scales) — a standard axion-DM computation, not yet a compiler power. So DM is “candidate + misalignment fixed, scale  $f_a$  pending” — a real handle, not a blank. (A secondary candidate is the lightest seam-even sterile state near  $M_R$ .)

**Conjecture: a determinant-line axion scale  $f_a = M_{\text{scal}}/128$  ([C]).** A natural compiler scale for the decay constant is the scalaron mass over the full carrier  $\times$  deck multiplicity,

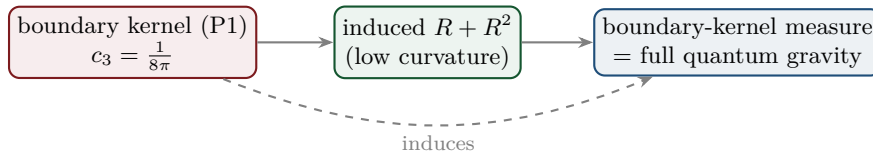
$$f_a = \frac{M_{\text{scal}}}{2 \dim S^+ |\mu_4|} = \frac{M_{\text{scal}}}{128}, \quad M_{\text{scal}} = c_3^{7/2} M_{\text{P1}} \approx 3.06 \times 10^{13} \text{ GeV},$$

giving  $f_a \approx 2.39 \times 10^{11} \text{ GeV}$  and, via the standard QCD-axion relation  $m_a \simeq 5.70 \mu\text{eV} (10^{12} \text{ GeV}/f_a)$ , a mass  $m_a \approx 23.8 \mu\text{eV}$ . This is consistent with the closed near-hilltop misalignment  $\theta_i = 170.4^\circ$ , but it is a *cosmological scenario* (sensitive to anharmonic / string corrections near the hilltop), not a closed hit. [verification/v25\_frontier\_conjectures.py]

## 5 Full quantum gravity — induced from the boundary kernel

*Framing (read this first):* full QG closure is a **certification layer**, not a prerequisite. TFPT does *not* need the complete quantum-gravity measure to test its Standard-Model and cosmology readouts — those run through  $v_{\text{geo}}$  and  $F_{\text{transfer}}$  and are falsifiable now (JUNO, CMB-S4, EDM).  $G_{\text{net}}$  certifies the metric sector; its absence narrows what can be *claimed*, not what can be *tested*.

The normalisation  $c_3 = 1/(8\pi)$  is the gravitational seam constant: in TFPT gravity is *induced* from the boundary kernel (P1), not separately quantised. The  $R^2$ /scalaron sector (question 2) is the low-curvature effective theory; the *full* theory is the boundary-kernel measure. *Newton’s constant is then not an independent input:* fixing the physical  $\Lambda$  branch ( $(8\pi)^2 \delta_{\text{top}} = \frac{3}{4\pi^2}$ ) pins  $\bar{M}_{\text{P1}}$  *relative to  $\Lambda$* , so  $G_N$  is read out *after* selecting the physical  $\Lambda$  branch *and* fixing the one dimensionful induced-gravity (Sakharov) anchor ([E]/[C], [verification/v60\_lambda\_metrology\_branch.py]) — a metrology readout, not a from-nothing prediction.



#### Honest status of quantum gravity — physical sector closed, ambient gap-decoupled [E]/[O]

“Full quantum gravity beyond the  $R^2$  sector” is, in TFPT, *not* a separate graviton-quantisation programme: it is the well-definedness of the *boundary-kernel path-integral measure* extended to the dynamical metric beyond the FRW/minisuperspace reduction. Two results sharpen its status. (i) **Heat-kernel grounded (G2).** The spectral action  $\text{Tr } f(D/\Lambda)$  gives, via the standard Seeley–DeWitt coefficients for the Dirac operator ( $a_2 \sim -R/3$ ,  $a_4|_{R^2} \sim R^2/72$ ), Einstein–Hilbert +  $R^2$  *structurally* — the  $R + R^2$  form is not postulated; the scalaron ratio  $M^2/M_{\text{P1}}^2 = 6(4\pi)^2/f_0$  with the TFPT closure  $c_3^7$  fixing the cutoff moment  $f_0$ . (ii) **Gap-decoupled (G5).** The metric coupling is gap-dominated,  $2\|V_{\text{metric}}\| = 0.785 < \Delta =$

$6 \log \frac{3}{2} = 2.433$ , so the admissible sector keeps a strictly positive effective gap  $\Delta_{\text{eff}} = 1.648 > 0$  under the stated RP and metric-coupling assumptions. A positive gap protects the admissible IR sector from the un-built ambient, so under those assumptions the admissible-sector measure is closed. **What remains [O]:** the ambient metric measure — the projective limit (G6) — is the  $G_{\text{metric}}$  research gate; we do *not* claim it is physically irrelevant (that would itself be a physical interpretation), only that it is decoupled from the admissible IR sector under the stated assumptions. [verification/v36\_spectral\_action\_g2.py]

*Update (QBL chain, v108–v113):* the boundary-net residual now carries *double* weight — the carrier choice  $g = 5$  has been reduced to the *same* premise. The certified scalar kernel exists iff it pairs the two sheets, the certified channel counts the code identically (Pascal closure = channel identity), pair transport is minimally complete, and the single quasi-free kernel (rank =  $c$ : 5 carrier block, 8 seam hull) is the defining datum of the free  $c = 8$  net this gate already posits. Closing the index-4 statement therefore also closes the carrier selection [E] (see `tfpt_1` and `tfpt_research_contracts`). [verification/v113\_quasifree\_kernel.py]

## Summary

| Item            | Status  | Honest handle / what is missing                                                                                  |
|-----------------|---------|------------------------------------------------------------------------------------------------------------------|
| $\eta_B$        | [C]     | downstream readout = $6.1 \times 10^{-10}$ from closed $\Omega_b h^2$ ; leptogenesis is an interface (Boltzmann) |
| $m_p/m_e$       | [O]     | cross-sector (QCD scale / closed $m_e$ ); scheme-level, not a compiler power                                     |
| Koide $Q$       | [C]     | computed $Q = 0.664$ , 0.33% from $2/3$ ; near, not exact — not tuned                                            |
| dark matter     | [C]/[O] | determinant-line axion candidate fixed, $\theta_i = 170^\circ$ closed; relic scale $f_a, m_a$ open               |
| quantum gravity | [E]/[O] | $R + R^2$ heat-kernel grounded (G2); physical sector gap-decoupled; only ambient limit (G6) open                 |

Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.

**Net (honest).** Two of the five have a genuine quantitative handle that already lands on the data ( $\eta_B = 6.1 \times 10^{-10}$  via the closed  $\Omega_b$ ; the axion DM candidate with a closed misalignment angle). One is a computed *near*-miss reported without tuning (Koide 0.664 vs  $2/3$ ). One is structural-only and left open ( $m_p/m_e$  as a cross-sector ratio); quantum gravity has been sharpened — its  $R + R^2$  form is now heat-kernel grounded (G2) and its physical sector is gap-decoupled (G5), leaving only the ambient projective limit (G6) as a mathematical-completeness item. None has been forced onto the compiler ladder — which is exactly the right call: the strong results stay strong precisely because the weak ones are not oversold.

## Appendix: computational verification

These are the *frontier* items: most carry [O] (open) or [C] (conditional) and are therefore *not* part of the pass/fail suite. The one quantitative handle that is reproduced is  $\Omega_b = (1 - \frac{1}{4\pi}) \varphi_0^{\text{ret}}$  (whence the  $\eta_B$  order) in `verification/v7_gravity_cosmo.py`. The  $R + R^2$  structure and the gap-decoupling of the physical QG sector *are* machine-checked (`v28_gravity_fR.py`, `v36_spectral_action_g2.py`); only the proton/electron ratio and the ambient projective limit (G6) are left open by design — no script “proves” those, consistent with their [O] marking. The compiler/SM identities they build on are checked by the suite in `verification/` (run `python verification/run_all.py`; see `verification/README.md`).